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Research on personalized distance education recommendation system based on deep learning

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Online learning environments must be personalized with intelligent systems that propose courses according to learners' individual preferences, activity patterns, and proficiency levels. Current recommendation models, however, fail to capture dynamic learner behavior and do not scale in big data. The research proposes a hybrid Putterfish Optimization Algorithm with Apriori (POA–Apriori)–Map-Reduce based Convolutional Neural Network (MR-CNN) methodology to address these issues, which integrates the Pufferfish Optimization Algorithm (POA) for global parameter tuning, the Apriori algorithm for association rule mining, and MapReduce-based Convolutional Neural Network (MR-CNN) for parallel deep learning. Data preprocessing was performed through normalization, feature encoding, and temporal smoothing and then model training with Udemy and Coursera datasets. The proposed model performed well with strong predictive performance of Precision@10 = 0.8769, Recall@10 = 0.8896, Normalized Discounted Cumulative Gain (NDCG)@10 = 0.8765, and lower error metrics with Mean Absolute Error (MAE)@10 = 0.2550 and Root Mean Squared Error (RMSE)@10 = 0.3568 compared to traditional models such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Deep Neural Networks (DNN) using both data sets. Statistical tests using paired t-tests ($p < 0.05$) confirmed the significance of these results. Experiments demonstrate that the hybrid POA–Apriori–MR-CNN approach significantly enhances recommendation accuracy, scalability, and robustness for massive-scale personalized education systems. The integration of optimization, association mining, and distributed deep learning marks a significant milestone towards adaptive and data-oriented e-learning environments.

Keywords Personalized education, Recommendation system, Deep learning, MapReduce, Interest mining

The educational system serves as a historical monument to the vast store of information and wisdom that exists in the cosmos. On the other hand, the "Gurukula" system[1] was well-known in ancient times for its students' constant access to instructors and wealth of information resources, as well as for the fact that they lived on the grounds while they studied. Over time, education has changed from strict, teacher-focused setups to more student-centered methods that focus on flexibility and involvement[2]. The historical development shows how education has broadened, but the main challenge today is making sure that learning systems can adjust to each student's unique preferences, speed, and situation. This issue is especially important now, as online education offers learners a vast array of course choices[3].

Personalized suggestions may be affected by language barriers, as the algorithm may encounter difficulties in accurately interpreting user preferences across different languages. The system's inability to accommodate the distinct preferences and learning needs of individual students may result in confusion during the selection of study materials. In conclusion, insufficient financial resources or inadequate infrastructure may hinder the system's ability to deliver high-quality, personalized recommendations. The Government of India has implemented various initiatives to tackle these issues and enhance educational standards nationwide. The organization has preserved its commitment to intellectual quality through various educational programs and institutions, even in contemporary contexts. The origins of the problem can be attributed to several factors, including an outdated curriculum, insufficient opportunities for hands-on learning, a shortage of qualified teachers, and additional related issues[4]. Digital technologies have clearly enhanced instructional effectiveness over the past five to ten years. Online education has increased the accessibility of high-quality education for all individuals. The initiative has achieved this by providing extensive opportunities for teaching and learning[5] and by significantly enhancing student learning outcomes, engagement, and pedagogical innovation.

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Despite the progress in online learning, learners still struggle to find courses that match their goals and skill levels. Traditional recommendation systems often don't adjust to individual learning styles or changing interests. This results in generic and unhelpful suggestions. This study tackles these issues by suggesting a hybrid deep learning-based personalized recommendation system. It models individual learner behavior and preferences in real time.

However, despite the rapid growth of online learning platforms like Coursera and Udemy, learners still struggle to find courses that truly fit their learning goals, interests, and abilities. Traditional recommendation algorithms, such as collaborative filtering and matrix factorization, have trouble adjusting to changing learner behaviors, sparse data, and large volumes of content. As a result, many online systems do not provide relevant or context-aware recommendations. This research tackles this issue by proposing a hybrid intelligent model that can capture learner preferences in real-time and improve the accuracy and scalability of recommendations. To overcome these challenges, this study combines the Pufferfish Optimization Algorithm (POA), the Apriori association rule algorithm, and a MapReduce-based Convolutional Neural Network (MR-CNN). Each technique contributes in a unique way: POA improves global optimization and parameter tuning, Apriori uncovers hidden connections between user interests and course features, and MR-CNN boosts feature extraction and large-scale data processing using distributed deep learning. Their combination creates a cohesive hybrid framework that balances accuracy, scalability, and ease of understanding for personalized course recommendations.

Recent studies highlight the increasing use of smart and adaptable systems in various fields. For instance, Falola et al.[6] show how AI-driven reminders enhance personalization and user engagement, while Wang et al.[7] showcase the effectiveness of deep learning and distributed computation for managing complex dynamic data. Inspired by these developments, this research employs hybrid deep-learning and optimization techniques for personalized distance education.

Contextualizing the evolution of education

This introductory section aims to establish a foundation by detailing the evolution of education throughout history. The analysis focuses on significant turning points, shifts in educational philosophy, and social impacts to offer readers a comprehensive understanding of the evolution of education over time. Comprehending the current situation and the challenges necessitating new approaches, such as personalized education, demands a foundation in the historical context.

The role of technology in educational

Transformation Technology has transformed the way information is learned, exchanged, and distributed; it is now an essential aspect of contemporary education. In this part, we will look at how technology has revolutionized teaching methods[8].

Emergence of personalized learning

The emergence of customized learning recognizes the uniqueness of each student, including their individual needs, preferences, and learning styles. This section of the introduction analyzes the reasons and factors that contribute to the emergence of customized learning. It explores the educational ideas that support this approach, highlighting the transition from uniform models to adaptive and personalized learning experiences. Comprehending the background and justification for personalized learning enables readers to see the imperative of using data-driven technologies such as recommendation systems[9].

In this research, the idea of differentiated teaching serves as the basis for the proposed system. Just as differentiated instruction adjusts teaching methods and materials to meet learners' needs, the proposed recommendation model uses data-driven algorithms to customize learning content. By analyzing learners' interests, performance, and engagement in real time, the system applies the principles of differentiated teaching in an intelligent, automated way.

The modification of instruction to meet the various requirements of students in mixed-ability courses while taking into account the variations in individuals' interests, learning preferences, and readiness levels is known as differentiated pedagogy. Every learner is regarded as an individual. To identify his or her talents, opportunities are provided for him or her to showcase their abilities. The type of differentiation a teacher chooses to use depends on the kids he teaches. Furthermore, teaching differentiation is a strategy that addresses each student's unique requirements and regards them as unique individuals rather than a set of rules to follow. To optimize learning chances in the classroom, the instructor adjusts both his teaching style and the learning exercises.

Learning processes, or how children learn, are the primary emphasis of differentiated pedagogy, rather than outcomes. While everyone is working toward the same objective, the instructor hopes to provide a framework that allows students some leeway to pursue their own learning paths[10]. Students gain confidence as they actively engage in the learning process and contribute in whatever manner they can. In groups, members contribute based on their abilities and areas of interest. On the other hand, when kids are allowed to make errors without facing consequences, it gives them a sense of acceptance, which encourages them to try new things. While doing so, a teacher who differentiates his lessons draws on his pupils' backgrounds and interests, promotes student independence while simultaneously strengthening group dynamics. He has faith in his pupils' abilities and the right to be unique. In the classroom, he also mediates and facilitates discussions. Students in similar settings report higher levels of enthusiasm and interest in learning, as well as a sense of trust in their teacher's ability to guide them and accept their mistakes[11–14].

The instructor should also think about the factors that contribute to his class's diversity, such as the parents' socioeconomic status and cultural background, as well as the students' perspective on school. On top of that, the educator works hard to establish solid rapport with each pupil. He goes out of his way to make sure they are treated like people, not just students. Additionally, he needs to be cognizant of the fact that students' needs go

beyond just studying; they also have an emotional need for connection, both with him and with one another. Last but not least, chances for serious conversation must be made available. A sense of community based on meritocracy and openness to new ideas should also be fostered[15,16]. In addition, it is important to highlight that learning to plan allows teachers to make more detailed decisions, like designing pedagogically oriented learning and teaching activity sequences and making appropriate use of available technologies and educational resources. Because of its importance in achieving success and enhancing the quality of education, the research community that has emerged around the topic has made it a priority to support teachers in their role as learning planners.

The principles of differentiated teaching form the basis for this research. They highlight the importance of focusing on each learner's needs and preferences when designing instruction. Applying these principles to digital settings needs smart systems that can understand and adjust to learner behaviors in real time. This study connects teaching theory with technology by creating a hybrid POA, Apriori, MR-CNN model that implements differentiation through data-driven personalization in online learning platforms.

Objectives and contributions of the study

The main objectives of this research are as follows:

- To develop a hybrid personalized recommendation model (POA-Apriori-MR-CNN) that combines optimization, association mining, and deep learning for distance education.
- To improve user interest mining through the POA-Apriori combination for precise identification of learner-course relationships.
- To create a distributed MR-CNN framework that can efficiently manage large-scale course and learner data.
- To assess the model using benchmark datasets from Udemy and Coursera, comparing its performance with existing top-K recommendation systems.
- To show improvements in precision, recall, MAE, RMSE, and NDCG performance, confirming the model's effectiveness for personalized e-learning.

Organization of the paper

The rest of this paper is organized as follows: Sect. "Related works" presents related works and research background. Sect. "Proposed recommendation system" provides a detailed explanation of the proposed hybrid POA-Apriori-MR-CNN model. Sect. "Experimental analysis" describes the experimental setup, dataset details, and performance evaluation. Finally, Sect. "Conclusion" wraps up the study and suggests directions for future research.

Related works

This study introduces a personalized course resource recommendation system utilizing deep learning, designed for implementation in online multi-modal multimedia education cloud platforms. The primary objective is to mitigate the problem of mistake propagation resulting from unstructured text within these systems. Before conducting an analysis of the semantic content of the text across different contexts, the BERT pre-training model is employed to derive the word vector from the course data set. Subsequently, the core information of the sentence is represented with greater precision through the long short-term memory network (LSTM), in which the multi-head attention layer allocates different weights to distinct word vectors. The process of personalized course resource selection involves the utilization of the CRF layer for the detection of sentence entities, followed by the application of the Sigmoid layer to extract relationships. This model demonstrates significant advancements over prior iterations. The experimental results indicate that the algorithm effectively recommends resources tailored to the courses of individual students[17].

The objective of this article is to analyze the challenges associated with the low computational efficiency and high propagation error rate in relation extraction and entity recognition. This research introduces the XMAMBLSTM framework, a deep learning-based approach designed to provide personalized recommendations for educational resources within the realm of intelligent education robots. XMAMBLSTM incorporates a multi-headed attention layer to facilitate weight distribution across individual word vectors, employs XLNet for the assignment of word vectors to text sequences, and integrates a Multi-Bi-LSTM layer to effectively represent the complex information inherent in the word vectors. The experimental findings indicate that the proposed technique exceeds the performance of the conventional collaborative filtering algorithm by over 5.05% and 17.3% across the two platforms, respectively, during the assessment of the intelligent teaching robot environment[18].

To manage the growing volume of user-generated content on the internet, there is a significant demand for advanced recommendation systems (RS). This study introduces a novel method that integrates sentiment analysis into recommendation system strategies, effectively addressing the challenges associated with conventional recommendation systems. The primary goal is to enhance the accuracy and reliability of user suggestions while simultaneously optimizing the performance of SA. The proposed framework consists of three primary components: sentiment analysis (SA), the integration of a recommendation system with sentiment analysis, and the development of personalized suggestions derived from the emotional context of user reviews, evaluated through the Bi-GRU approach. Comprehensive evaluations conducted on a range of musical instruments indicate significant enhancements in performance metrics. The Bi-GRU approach achieves an accuracy rate of 89%. The Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) for user suggestions based on Simulated Annealing (SA) are 1.83 and 1.71, respectively. The RMSE is 2.18, and the MAE is 2.00 when employing item-based suggestions. This method represents a significant advancement, enabling the provision of more accurate and reliable suggestions to users, thereby enhancing their satisfaction and overall experience quality[19].

The proliferation of courses available on these platforms has led to a significant increase in the popularity of online education in recent years. Learners may encounter difficulties in identifying the most relevant and

suitable courses because of the extensive array of available options. A strategy utilizing deep learning is proposed to enhance the accuracy and relevance of personalized course recommendations through the application of user data and sentence embeddings. This will assist in addressing this challenge. The technique effectively generates dense vector representations of course material utilizing the Multilingual Universal Sentence Encoder (MUSE) to capture semantic links across courses. Unique characteristics of users, including education level, birthday, and location, are incorporated into these embeddings to create comprehensive profiles. To identify the courses that best match each user, we employ two advanced recommendation models: GRU4Rec and Convolutional Sequence Embedding, both of which were trained on the edX dataset. The results indicate improved model performance, resulting in more accurate and customized course recommendations for users[20].

This research presents a new methodology for the optimization of teaching resources and the adjustment of students' learning trajectories through the integration of advanced reinforcement learning technologies. The Actor-Critic algorithm facilitates efficient resource allocation and provides accurate recommendations for individualized student learning routes by enabling immediate strategy updates. Additionally, the decision-making advantages of DQN are leveraged in high-dimensional state spaces. This study achieves rapid strategy iteration through the utilization of the strategy network (Actor) for exploring and designing the learning route strategy, while employing the value network (Critic) to assess the quality of the strategy. The Q-value function is constructed utilizing a deep neural network to enhance the effectiveness and enjoyment of student learning. This network is utilized to select the optimal or near-optimal scheme for allocating learning resources from a vast array of feasible actions. Enhancing the efficiency of practical course administration provided a new perspective and a feasible strategy for the progression of intelligent education moving forward[21].

The field of education has transitioned into a new phase characterized by personalized learning, driven by the ongoing advancements in artificial intelligence technologies. This research focuses on the development of a personalized education recommendation algorithm utilizing artificial intelligence. The objective is to enhance the quality of education recommendation services available to students through the application of advanced AI technologies. This document initiates a comprehensive analysis of the challenges associated with traditional educational paradigms, including the generalized, uniform teaching methodologies, the segmentation of subject content, and the neglect of students' unique interests. It is recommended to implement a personalized education recommendation system to address these issues and effectively accommodate the diverse learning requirements of students. To construct a smart education recommendation system, artificial intelligence technologies, such as deep learning, have been utilized alongside big data analysis. The system employs recommendation algorithms to generate individualized lesson plans and propose resources tailored to each student. This customization is based on various factors, including student interests, historical performance, and preferred learning styles, as well as other multi-dimensional data points. The trial results indicate that the AI-driven personalized education recommendation system has markedly enhanced students' learning outcomes and increased their engagement in academic activities. Students indicated an increase in engagement, improved academic performance, and enhanced overall satisfaction with their educational experience following the implementation of the personalized suggestion system. The recommendation system possesses the capability to adapt to the evolving interests and learning styles of students due to its advanced intelligence, allowing for continuous learning and optimization[22].

A customized approach for suggesting higher education options to students is significantly dependent on the structure of the learner model. At present, the majority of learner models may benefit from a more scientific approach; these models often use a single, computationally efficient way to gather dimensions and feature properties. Accordingly, a Multivariate Hybrid Criteria Fuzzy Algorithm (MHCFA) for tailored college suggestion was presented in this research. The MHCFA undergoes its training using SFAT, which stands for Social Optimization Algorithm as well as Feedback Artificial Tree. The material of higher education is also grouped using Deep Fuzzy Clustering (DFC). To choose the most appropriate material, the RV-Coefficient is used. In addition, the characteristic is retrieved using numerical and All Caps data for additional tailored suggestions. And when it comes to Precision, Recall, and F-Measure, the data reveal that SFAT_MHCFA really shines[23].

With the tourist industry's rapid expansion in recent years, creating a new educational paradigm to train future professionals has emerged as a top priority, and the OBE model has emerged as a frontrunner in this movement. This paper presents a program design for an OBE-based tourism management program that takes into account two factors: the learners' personality traits and their learning resources. It then builds a model of the learners' personality traits using criteria like style similarity, behavioral similarity, and the learner influence relationship. Finally, it suggests a deep learning-based personality recommendation method, with the experimental class and the control class serving as the experimental objects, respectively. The authors conducted an analysis of the students' cognitive levels, learning styles, and interaction behavior characteristics to validate the model and the program design. The experimental group recorded a higher average score of 84.91, in contrast to the control group's average score of 83.63. The OBE education model, which is grounded in deep learning principles, has the potential to improve students' learning performance[24].

This research aims to develop a recommendation model for personalized resources in early childhood education by employing deep learning techniques and methodologies associated with recommendation systems. One of the challenges associated with tailored early childhood education resource suggestion involves managing features that are both limited in availability and high-dimensional in nature. This study utilizes the DeepFM model to address the identified challenge. The DeepFM model utilizes feature embedding techniques to convert high-dimensional discrete features into a dense vector space characterized by reduced dimensions, effectively addressing the challenges associated with high-dimensional data. Our research aims to enhance the DeepFM model by considering the influence of cross-features and integrating sequential data related to user behavior. The objective of this innovation is to analyze the interaction links between characteristics and high-order feature combinations. It aims to accurately identify young children's interests and preferences, providing

them with tailored suggestions. This project aims to enhance early childhood education by developing a model that provides tailored educational resources for each child, addressing their individual needs. The references provided by this study will significantly enhance research and practice in the domain of tailored early childhood education advice[25].

To improve students' learning experiences, it is true that several education platforms have introduced individualized learning services. Instead of a service that filters and suggests classes based on basic information about students and courses, we need one that suggests personalized classes according to students' past learning experiences. Based on the findings of analyzing large data collected from a real-time education platform, we provide recommendation indicators in this study. In addition, we provide a machine learning-based recommendation system that matches students with appropriate classes. In the long run, this platform may optimize learning synergy and student successes by offering a recommendation service that is student-centered[26].

Online education has exploded in popularity in the information era due to the proliferation of high-quality educational resources made possible by the proliferation of high-speed Internet and other forms of education informatization. By enhancing the conventional co-recommendation algorithm and introducing hybrid suggestions, this research builds a customized system and co-recommendation technology for English language teaching materials. In order to compare the four systems' efficacy, the system's performance is tested empirically. Outperforming the other three recommendation systems across all datasets, the enhanced algorithm emerges as the clear winner. With a T-value of $1.81 > 1.645$, the experimental class aided by the hybrid recommendation system in college and university English instruction achieved an average grade mean of 27.54, outperforming the comparison class's 25.33. Validity and stability in performance and practical application are good features of the upgraded personalized recommender system[27].

Students have a dilemma when deciding which online courses to register in due to the sheer number of options. Our innovative recommendation system, BERT (Bidirectional Encoder Representation from Transformers), proposes a solution to this problem. Learners' profiles and enrollment history are used to develop tailored suggestions, which boost engagement. Our research takes a look at 3678 course records from Udemy in four different categories, including material for students of varying skill levels. The technology, which makes use of BERT, claims to transform online education by guaranteeing the best possible course selection. Its performance and usability were highlighted by validation through a comprehensive survey, which offered ideas for improving online education platforms and empowering learners[28].

Online education has grown in popularity thanks to the Internet, and with the proliferation of educational materials comes a greater need for tailored recommendation systems. This research addresses this problem by suggesting an optimal collaborative filtering algorithm-based customized recommendation system. Collaborative filtering algorithms are commonly employed in recommendation models because of the robust user involvement they provide. But flaws like cold start might reduce the model's efficiency. In order to combat this issue, this research presents content recommendation systems. To successfully accomplish individualized suggestions, a hybrid recommendation model may be built on top of the two methods. At the same time, this research zeroes down on the system's critical modules, employs standardization and dimensionality reduction procedures to further lessen the computing load, and so on. In the end, the study compared the model against others to make sure it was reliable. Compared to the utility recommendation model (2.68% greater accuracy) and the rule recommendation model (7.99% higher accuracy), the mixed recommendation model performed better in the experiments. The proposed customized recommendation approach in the study utilizes an enhanced collaborative filtering algorithm, demonstrating effective functionality[29].

This research study examines the performance assessment of user-based collaborative filtering and content-based recommendation strategies, aimed at developing advanced recommender systems tailored for students majoring in information systems. This study combines the primary dataset gathered from the Knowledge-Skill-Attitude framework with data from the Faculty of Information Systems at the University of Economics and Law to assess the effectiveness of these two models in the development of personalized recommendation systems. The Collaborative Filtering and Content-Based models underwent empirical evaluation using key metrics such as F1-Score, Precision, and Recall. This assessment aimed to deliver a comprehensive understanding of their effectiveness in generating a Recommendation System for the University of Economics and Law. The results indicate that the Collaborative Filtering method achieves a high score in the Precision category. The Content-based approach demonstrates enhanced memory skills, potentially making it appropriate for addressing diverse student needs within the classroom environment. Recommendation systems are transforming higher education through personalized learning experiences and curriculum alignment with industry objectives, as demonstrated in this article. Future research should explore a hybrid approach that integrates the functionalities of both Collaborative Filtering and Content-based methods. This approach will assist in addressing the limitations inherent to each model, as each possesses a distinct set of constraints. The findings provide a foundation for additional research on hybrid recommendation methods, which enhance the efficiency and personalization of the learning experience for students. Furthermore, they offer actionable insights for educational institutions and organizations to refine educational content and career development resources[30].

An intelligent recommendation system can enhance Photoshop (PS) classes by integrating political and ideological teaching materials. The system comprises components such as learning management system (LMS) integration, data collection and analysis, algorithm development, and system design. To enhance the relevance of ideological and political material in Photoshop lessons for students, data was gathered regarding their interests and learning habits. An algorithm was developed to enhance the efficacy and accuracy of recommendations through the integration of content-based filtering with collaborative filtering and additional methodologies. A robust backend architecture managed data processing and distribution, whereas an intuitive frontend interface facilitated user interaction. Integration with LMS ensures compatibility and enhances user experience. The

intelligent recommendation system leverages these components to provide customized and impactful content suggestions, thereby improving Photoshop lessons and fostering political and ideological awareness[31].

Recent work on cloud-based learning equalization highlights the importance of data interoperability. For instance, Surendran et al.[32] created a scalable education-cloud model at the University of Essex. This model balances course delivery and computing resources across institutions, showing how edge-cloud integration can lead to fair access to digital learning content. Security and trust in academic credentialing improved with the Secured Accreditation and Equivalency Certification using Merkle Mountain Range and Transformer-Based Deep Learning Model for the Education Ecosystem (Krishnan et al.[33]). This model combines blockchain with transformer networks to ensure authenticity, privacy, and verifiable equivalency in online education platforms.

Gamified and inclusive learning has also gained from artificial intelligence frameworks. The study Game Therapy for Specially-Abled Individuals with PPO Reinforcement Learning in VR-Based Educational Games (Joypriyanka et al.[34]) introduced a VR-based system that customizes rehabilitation and learning pathways through reinforcement learning and immersive interfaces. Additionally, the Impact of Equivalence Assessment in the Education Sector using the XAI Model of Blockchain with ECTS (Krishnan and Surendran[35]) emphasized the importance of explainable AI-driven blockchain frameworks for assessing course-credit equivalency. This approach increases transparency and accountability in higher education recommendation and accreditation systems.

Research Gap Summary: Although many models[17–31] have improved personalized education through deep learning, cloud, and blockchain integration, several challenges remain. Current systems struggle with interoperability across different learning platforms, have limited flexibility to changing learner behavior, and seldom offer explainability or credential security. The proposed hybrid POA-Apriori-MR-CNN framework addresses these issues by combining optimization, distributed computing, and smart analytics to provide scalable, clear, and adaptable course recommendations (Table 1).

Advances in recent social network analysis also offer useful insights for recommendation and personalized learning systems. For example, graph-based community detection and semi-supervised deep learning methods have been used to analyze structural relationships between users and content. Semi-supervised and Deep Learning Approaches to Social Network Community Analysis discusses hybrid models integrating labeled and unlabeled data to determine user communities efficiently (De Santo[36]). In the same vein, Graph Clustering Techniques for Community Detection in Social Networks offers graph-based clustering methods that reveal latent patterns from massive-scale networks Daneshfar et al.[37]). Conceptually, these are in sync with our work since both contexts focus on finding useful associations—in our work, between learning individuals and learning materials—using deep learning and optimization-based models.

Proposed recommendation system

Background overview

Foundations of personalized learning

The principles of customized learning establish a foundation for a transformative era in education, characterized by data-driven methodologies that prioritize the specific needs of individual students. This article examines the key aspects of customized learning, detailing its definition, significance, and the transformative shift towards personalized learning experiences.

Defining personalized learning in education

Personalized learning is an instructional approach designed to address the individual needs of each student through the analysis of their historical performance, current interests, and preferred learning modalities. Students exhibit differences in learning speeds, strengths, areas for improvement, and preferred methods for absorbing new material; this principle is essential to individualized learning. The objective of customized learning is to deliver an education tailored to individual students by integrating data analytics, technology, and adaptable teaching methods, in contrast to the traditional, uniform approach. The primary objective of customized learning is to identify and leverage the distinct talents of each student. This may involve the presentation of diverse learning materials, adjustments to the pace of instruction, or modifications to the methods of information delivery to accommodate different learning preferences. To enhance student engagement, motivation, and overall academic achievement, educators aim to tailor the learning process.

Year	Method/model	Key features	Strengths	Limitations	Reference
2021	Education cloud equalizing service	Cloud-based resource equalization	Scalable & equitable access	Limited personalization	Surendran et al
2023	BERT-LSTM semantic extraction	Multi-modal course data	High semantic accuracy	High computation cost	Zhang et al
2024	PPO reinforcement learning VR games	Gamified learning for special needs	Adaptive engagement	Narrow learner group	Joypriyanka et al
2024	XAI Blockchain ECTS Model	Explainable credit assessment	Transparent decisions	Requires high blockchain cost	Krishnan and Surendran
2025	Blockchain + Transformer accreditation	Deep learning with Merkle Range	Secure credentialing	Implementation overhead	Krishnan et al
2024	MR-CNN recommendation system (proposed)	Hybrid MapReduce–CNN	High accuracy & scalability	Complexity in tuning	*This work*

Table 1. Summary of recent approaches in personalized education and recommendation systems.

Importance of addressing individual student needs

In the framework of personalized education, it is essential to address the distinct needs of each student. Each classroom exhibits distinct characteristics, as every student contributes a unique combination of experiences, knowledge, and learning methodologies. To ensure that all students receive appropriate support, personalized learning considers and addresses these variations. Educators can enhance classroom climate by addressing the unique needs of students through the implementation of individualized lesson plans. By adopting this approach, educators can identify the specific challenges faced by each student and promptly adjust their instructional methods as needed. Customized learning promotes a sense of empowerment among students by identifying and addressing their individual talents and challenges.

The shift towards tailored educational experiences

Recent advancements in technology and an enhanced understanding of effective pedagogy are driving a significant transformation in the educational landscape, focusing on personalized learning experiences. The adoption of a more flexible and adaptable approach in personalized learning represents a departure from traditional, rigid educational methods. The integration of data-driven insights to inform pedagogical methods is fundamental to this transformation. Determining each student's strengths, weaknesses, interests, and areas for improvement is essential. Educational technology, including learning analytics and recommendation systems, plays a critical role in this process. Educators may optimize their students' educational journeys and achieve better results by adopting this data-driven approach and designing adaptable learning environments that change with each student's requirements.

Recommendation system model

True customized suggestion is not possible in this experiment due to the high data analysis costs, difficulty in generating association rules, and poor accuracy of the teaching intelligent system's personalized recommendation model of association rules. In order to accomplish the goal of providing students with individualized recommendations of course resources, it is essential to construct a recommendation system in accordance with the collaborative filtering personalized recommendation model. Figure 1 depicts the suggested model.

Figure 1 shows the complete workflow of the personalized recommendation model. The process starts with collecting learner-course data, followed by preprocessing and removing noise. The hybrid POA-Apriori module identifies interest associations, while the MR-CNN module carries out feature extraction and distributed learning. Finally, ranked course recommendations are produced and presented to users through the learning platform interface.

Personalized learning uses data analytics and smart systems to adjust educational content for individual learners. It recognizes that learners progress at different speeds, have varied interests, and prefer different learning styles. To address these differences, systems must be able to model learner behavior dynamically and offer suitable recommendations. Building on this foundation, the proposed hybrid POA, Apriori, MR-CNN model uses deep learning and optimization techniques to achieve personalization in real time. This shift from theoretical personalization to practical implementation is the basis for the system described in the following section.

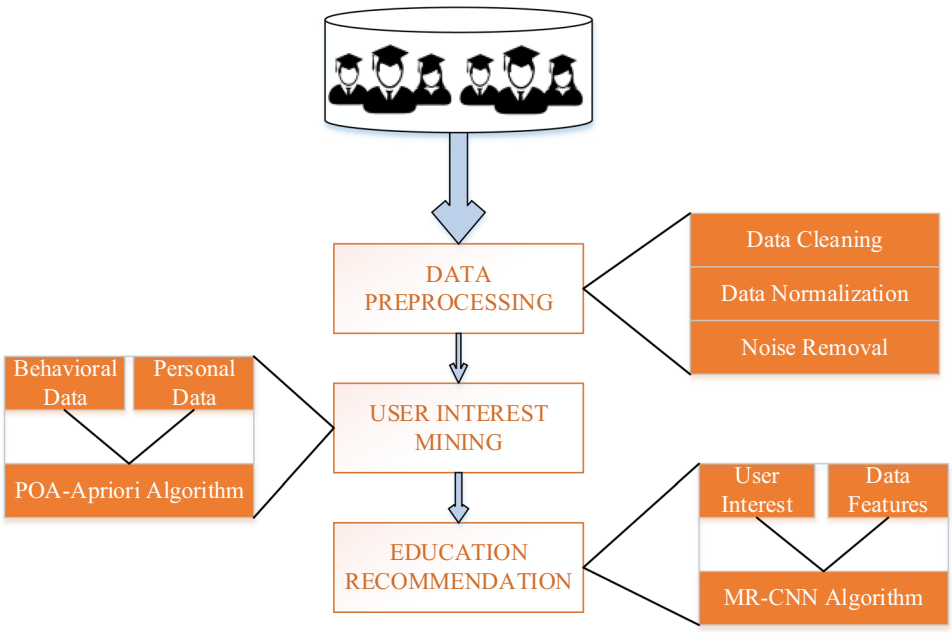


Fig. 1. Proposed recommendation model.

Data preprocessing

In order to solve any data analysis challenge, data must first be preprocessed. The reliability of the data is the primary determinant of the model's precision. Information transformations, data purification, and the identification and treatment of missing values are standard components of every data preparation procedure. To train a machine-learning algorithm, you must first prepare the data. There are properties upon which the mathematical model rests. As an example, in a survey, the characteristics stand in for the questions and intercepted keywords, while the values of the attributes represent the alternatives and important attitudes related to those questions. To create an N-dimensional space, we next specify each characteristic as a spatial dimension. An $M * N$ matrix, where M is the maximum number of topic possibilities, and an element is each person's decision in each dimension, represents their information. Vectors and matrices are the element kinds; "1" indicates selection and "0" indicates unselection. N components make up each person's information. One of the data sources, including multiple-choice, single-choice, and sorting questions. Table 2 has three questions, indicating the presence of three dimensions. In the second MVC, you'll need to choose ACE, which is a vector that's also set to 1 at that exact spot. Due to the sequential nature of the sorting problems (presuming CBADEFIHG as the order), the track sorting problem differs differently in representation from the previous two questions. Dimension enhancement processing, denoted by a matrix, is an essential operation. The first choice in the sorting issue is depicted in the first column of the matrix. Here, C, the equivalent position is 1. There are two possible solutions to the sorting problem, and the second column shows them. Here, B stands for 1, and the same holds true for the other positions. Next, let's pretend that the N-question questionnaire data is represented by a matrix with dimensions M N × 1 and N N 1 ≥ (where N1 is the matrix that remains after adding dimensions). The information from K's N-question survey may be represented by K matrices of dimensions M N × 1.

Data normalization

Data mining makes use of data normalization as a method to standardize dataset values. Data normalization is crucial since many ML algorithms output superior results depending on the size of the input characteristics. When using the Z-Score method, feature values are normalized to a mean of zero and a standard deviation of one. You may achieve this by taking each value and removing the feature's mean, and then dividing the result by the standard deviation. Using the data set A's mean and standard deviation, this method normalizes the values. The normalization process follows Eq. (1), where each feature value is standardized using the mean and standard deviation.

$$v' = \frac{v - \overline{A}}{\sigma_A} \tag{1}$$

$v\prime, v$ is the new and old of each entry in data correspondingly. σ_A, A is the standard deviation and mean of A respectively.

Noise removal

The term "noisy data" describes information that contains a great deal of extraneous, irrelevant details. The phrase is frequently used interchangeably with corrupt data and encompasses data corruption as well. Data that a user system is unable to properly comprehend and interpret is also included. Unstructured text, for instance, is incompatible with many systems. Improper handling of noisy data can distort outcomes and negatively impact data analysis. It is possible to extract useful information from noisy datasets by employing statistical analysis. There are several technical uses for Kalman filtering. Modeling the way the central nervous system controls movement is another area where Kalman filtering excels. There are two stages to the algorithm's operation: prediction and updating. The Kalman filter produces estimates of state variables along with their associated uncertainty in the prediction phase. Estimates are updated through a weighted average approach, assigning greater significance to those estimates that exhibit higher certainty. This process occurs upon the observation of subsequent measurements, which are inherently subject to various errors, including random noise. This program contains a self-referential loop. For real-time operation, additional historical data is unnecessary; only the current input measurements, the previously computed state, and the associated uncertainty matrix are required.

Assuming that the actual state has evolved from the state at time in accordance with the Kalman filter model, which is shown in Eq. (2)

$$x_k = F_k x_{k-1} + B_k u_k + w_k \tag{2}$$

Parameter	Specification/size
Operating system	Windows 10
Central processing unit	Intel 5-8400
Graphic processing unit	NVIDIA
Memory	32 GB
DL framework	TensorFlow
Programming language	python3.6
Number of hidden layer nodes for G and D	(50,100,150,200,300)

Table 2. Experimental environment and experimental parameter settings of model validation.

x_k: system state vector (engagement features); F_k: state-transition matrix; B_k: control-input matrix; u_k: control vector; w_k: process noise ~ N(0,Qk); z_k: measurement vector; H_k: observation matrix; v_k: measurement noise ~ N(0,Rk); P_k: state-error covariance. Typical values: Qk = 10^{-3}I; RkR_kRk = feature-wise variance; decay α = 0.95. The process noise, w_k, is taken to be distributed from a zero-mean multivariate normal distribution. Using the coefficient of determination is depicted in Eq. (3),

$$Q_k : w_k \sim N(0, Q_k) \tag{3}$$

A measurement of the actual condition x_k is taken at time k using the Eq. (5),

$$z_k = H_k x_k + v_k \tag{4}$$

The observation model, denoted as H_k, transforms the real state space into the observed space.

We presume that v_k, the observation noise, is zero-mean Gaussian white noise with a covariance as shown in Eq. (5).

$$R_k : v_k \sim N(0, R_k) \tag{5}$$

Preprocessing Workflow: Text fields like course titles and descriptions were tokenized and converted into TF-IDF and word-embedding forms. Numerical features, including ratings, enrollments, and price were normalized using Z-score scaling, while categorical features such as subject area and instructor were one-hot encoded. Missing values were filled with median or mode values. Then, the dataset was randomly divided into 80% training and 20% testing subsets. The Kalman filter was used solely to stabilize temporal fluctuations in learners' engagement time series, like daily logins, ensuring accurate trend estimation without changing static course attributes.

User interest mining: POA-Apriori

The pufferfish belongs to the Tetraodontiformes family and is classified as an estuary fish. This fish resembles the porcupinefish, known for their large spines. Pufferfish exhibit a notable vulnerability due to their limited mobility. A distinctive method of self-defense employed by pufferfish involves the expansion of their elastic stomachs, transforming them into a large, spiky sphere to deter predators. In the selection of CHs, we employed the Pufferfish Optimization Algorithm (POA).

Mathematical model of POA

Potentially useful for resolving optimization problems, the proposed POA approach is an iterative population-based search tool. In order to investigate potential solutions, it mobilizes the population's power. In order to determine the values of the problem's decision variables, each POA member takes into account its position in the search space. You may think of the POA as a collection of possible answers to the problem; each answer might be a mathematical vector with a decision variable at its core. The total number of POA members is used as the population in the method. From a mathematical point of view, the community of these vectors may be described using a matrix. Equations (6 & 7) are utilized to determine the primary role of every POA member at the commencement of the operation,

$$X = \begin{matrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_N \end{matrix} \tag{6}$$

$$x_{i,d} = lb_d + r \cdot (ub_d - lb_d) \tag{7}$$

In this context, X represents the POA population matrix, where X_i denotes the ith POA member (potential solution), and x_(i,d) indicates its dth dimension within the search space (decision variable). The variable N signifies the population size, m refers to the number of decision variables, r is a random integer ranging from 0 to 1, and ub_d and lb_d represent the upper and lower bounds of the dth decision-making variable, respectively.

The assessment of the objective function related to the problem can be conducted with each member of the POA considered as a possible solution. A vector can represent the evaluated values for the objective function of the problem, as outlined in the corresponding Eq. (8).

$$F = \begin{matrix} F_1 \\ \vdots \\ F_i \\ \vdots \\ F_N \end{matrix} \tag{8}$$

F is a vector representing the evaluated objective function, and F_i is the ith element of the POA's objective function. In this case, we may express F_i as shown in Eq. (9):

$$F_i = \{EL_i, Mob_i AD\} \tag{9}$$

Three variables are defined here: energy level (EL), mobility (Mob), and average distance (AD) to neighboring nodes.

The quality of the solutions given by each POA member may be evaluated using the evaluated values of the objective function. An element is deemed optimal if and only if its best evaluated value is equal to the objective function, and worst if and only if its worst evaluated value is equal to the objective function. With every iteration, the position of the POA members inside the problem-solving area is updated. Changing the best member based on this comparison should also be part of each iteration, as the target function is assessed with fresh data.

The suggested POA method readjusts the population's problem-solving space location by simulating pufferfish interactions with their predators in the environment. In this specific life cycle, pufferfish are the main prey. After this, the pufferfish will employ its defensive strategy, which entails transforming into a spherical form adorned with pointed spines, to instill fear in the predator and ultimately allow it to escape. Consequently, there are two steps involved in updating the location of POA population members every iteration: (i) exploration, which indicates that the pufferfish is under assault, and (ii) exploitation, which suggests that the pufferfish is attempting to protect itself.

Phase 1: Predator Attack towards Pufferfish (Exploration Phase).

First, POA updates the population's location using the pufferfish simulated predator assault approach. Pufferfish are easy prey for predators due to their sluggish swimming pace. To keep the POA members updated on the newest happenings in the sphere of problem-solving, we imitate the predator's position shift when it pursues the pufferfish. As the pufferfish's predator draws near, the POA members' positions undergo dramatic shifts, enhancing the algorithm's capacity for a worldwide search.

In a population-oriented architecture (POA), all of the pufferfish fight for the same prey. The pufferfish whose goal function is most highly valued will attack where its peers are most highly valued. Using this Eq. (10), we can classify all pufferfish into one of several groups,

$$CP_i = X_k : F_k \tag{10}$$

The collection of potential pufferfish homes for the ith predator is denoted as CP_i. The population member X_k who maximizes the objective function better than the kth predator is denoted as F_k.

The POA design was based on the premise that the predator would randomly pick one pufferfish, known as the chosen pufferfish (SP), from the candidate set in the CP set. Based on the predator's approach model to the pufferfish, each member of the POA moves in the problem-solving space as described by Eq. (5). Then, according to the Eqs. (11) and (12), the old one is substituted with the new one if the new position increases the target function value.

$$x_{i,j}^{P1} = x_{i,j} + r_{i,j} \cdot (SP_{i,j} - I_{i,j}.x_{i,j}) \tag{11}$$

$$X_i = \begin{cases} X_i^P F_i^{P1} \leq F_i \\ X_i else \end{cases} \tag{12}$$

Phase 2: Defense Mechanism of Pufferfish against Predators (Exploitation Phase).

Phase two of AI-based population optimization entails relocating the population in response to a predator attack modelled after the pufferfish's defense mechanism. Pufferfish are able to outsmart predators by encasing themselves in a ball of spines after filling their extremely elastic bellies with water. The predator doesn't take advantage of the pufferfish's weakness; it runs away in terror. In order to enhance the algorithm's ability to utilize local search, we simulate the predator relocating away from the pufferfish, leading to minute changes in the locations of the POA members.

By simulating the predator's position change as it retreats, we may update the equation to find each member of the POA at a different place. This new position is used to replace the associated member if it raises the value of the objective function, according to Eq. (8).

Since the algorithm has been the target of algorithm improvement efforts in POA design, Eq. (8) will be utilized. We may test if computing a POA member's new location improves the solution by comparing the values of the objective functions. In the first scenario, the appropriate POA member can choose to take a different approach; but, in the second, sticking with their old position will provide a weaker solution. Therefore, as can be seen in Eqs. (13) and (14), the process of updating each POA member is contingent upon the value of the goal function increasing.

$$x_{i,j}^{P2} = x_{i,j} + 1 - 2r_{i,j}.\frac{ub_j - lb_j}{t} \tag{13}$$

$$X_i = \begin{cases} X_i^{P2} F_i^{P2} \leq F_i \\ X_i else \end{cases} \tag{14}$$

As shown in Fig. 2, the POA–Apriori algorithm flowchart demonstrates the iterative optimization process Updating the positions of all POA members according to the exploration and exploitation stages is the first step after each algorithm iteration. Equations (4)–(8) are used to update the locations of POA members until the algorithm reaches its last iteration, following which the next iteration starts. After comparing the evaluated values for the objective function, the position of the best POA member is updated and saved with each iteration. After the method is fully implemented, the optimal POA member's position is proposed as a solution.

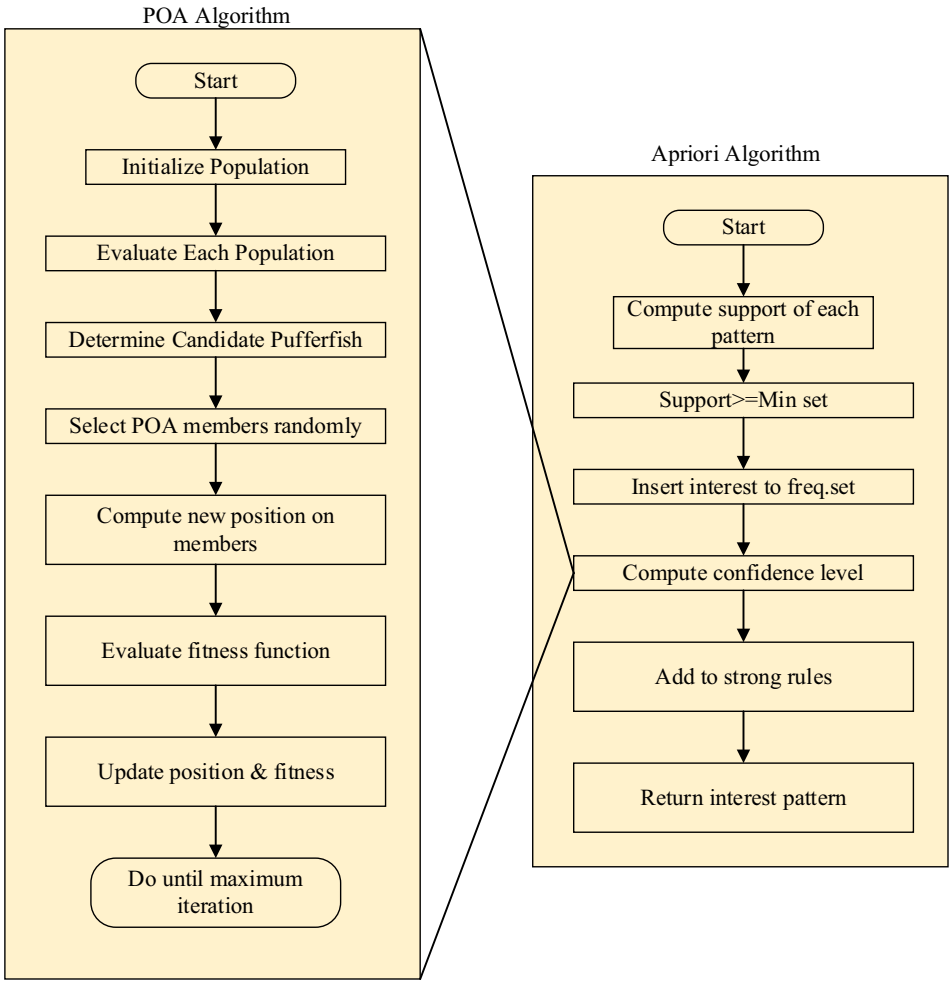


Fig. 2. Flowchart of proposed hybrid POA-Apriori algorithm.

The Apriori algorithm was used to identify frequent associations between learners and courses using support and confidence thresholds. However, choosing the best thresholds manually can lead to repeated or incomplete association rules. The Pufferfish Optimization Algorithm (POA) improves Apriori by automatically adjusting these thresholds through a global search. Each POA individual represents a candidate pair of support and confidence values. During each iteration, exploration and exploitation phases modify these parameters to enhance rule relevance and reduce redundancy. Once the process converges, Apriori applies the optimized thresholds to derive the most significant association rules. Figure 2 shows this workflow—from learner and course data input, POA setup, iterative parameter changes, to Apriori rule creation and the final mapping of user interests used for course recommendations.

Mathematical model of Apriori

Using relational databases, the Apriori algorithm may learn association rules and frequently used item sets. It works by finding the most common things in the database and building on them to create bigger and bigger groupings of objects as long as those sets appear frequently enough. The Apriori algorithm employs a "bottom up" methodology, wherein it iteratively evaluates candidate groups against the dataset by expanding frequent subgroups incrementally, one item at a time, a process referred to as candidate generation. The algorithm concludes when no additional successful extensions are identified.

Apriori utilizes a breadth-first search method for efficient candidate item set counting, implementing a Hash tree structure. Item sets of length k-1 are utilized to construct candidate item sets of length k. The candidates exhibiting an infrequent sub-pattern are subsequently eliminated. The candidate set includes all sets of items with k elements, as stated in the downward closure lemma. The next step is for it to search the transaction database for groups of applicants that often purchase certain items.

The following presents the pseudo code for the algorithm applied to a transaction database T, utilizing a support threshold ε. T represents a multiset, while C_k denotes the candidate set for level k. Standard set theory notation is applied otherwise. According to the downward closure lemma, the count [c] accesses a field of the data structure representing candidate set c, which is initially set to zero. At each stage, the algorithm is designed to generate the candidate sets from the large item sets of the previous level. The following discussion

omits numerous specifics; however, the data structure responsible for storing candidate sets and counting their frequencies is generally the most critical element of the implementation.

MapReduce DL based recommendation model

MapReduce is a distributed computing architecture, initially proposed by Google, recognized for its ability to process large volumes of data across extensive clusters comprising thousands of cost-effective computers. It is utilized extensively in large data processing applications. With MapReduce, processing, reading, and writing to big data sets can all be done simultaneously. First, the user defines a Map function to process the key-value data set. Then, they run the intermediate set through the Reduce function. Lastly, they combine all the values of the same key. This sets of key-value pairs is called an intermediate set. The open-source MapReduce implementation known as HadoopMapReduce is utilized in this project.

Data processing and job distribution among nodes are both handled by MapReduce. Two separate components, a mapper and a reducer, make up every MapReduce task. A collection of key-value pairs is passed into the mapper as input, and the data is transformed based on the logic provided in the map function. The reducer receives its input from the new collection of key/value pairs that it produces. In order to get an aggregated output, the reducer takes key-value pairs and merges them using the key.

CNN

Both regular neural networks and convolutional neural networks follow a similar pattern of operation: Initially, a linear transformation is computed between the input and the weights. Subsequently, a non-linear activation function is applied. Following this, the error is calculated. Finally, the error is backpropagated to the weights. The distinction between the two systems lies in the connections formed between their respective networks of neurons.

In neural networks, each neuron within a layer operates independently and does not share connections. Instead, it establishes connections with every neuron in the following layerz . Overfitting happens when a trained model aligns too closely with the noise present in the data, which negatively impacts its capacity to generalize effectively to new instances. Fortunately, CNNs excel in processing images by employing convolution and local receptive fields, which effectively manage the number of parameters involved. Consider employing a 5×5 filter, systematically moving it across the entirety of an image to identify features that correspond with the filter. This describes the precise method of performing convolution, where the filter is initialized with weights, and dot products are calculated between these weights and the input image, mirroring the process used in neural networks. In convolutional structures, neurons are not interconnected in a fully connected fashion. Instead, each neuron in a given layer connects only to a localized area of the preceding layer, a relationship determined according to the dimensions of the filters or receptive field.

The dimensions of the receptive field determine the spatial connectivity of neurons to their inputs, affecting both the width and height, while uniformly covering the entire depth. Applying a receptive field with dimensions of $5x5$ to an image of size $300x300x3$ yields neurons that possess $5 * 5 * 3 = 75$ weights. A wide receptive field can include a greater amount of contextual information; however, it may lead to a reduced perception of finer details. Secondly, it is noted that in Fig. 2, the border of the input is encircled by zeros. This mechanism is engineered to preserve the dimensions of both the input and output, thereby preventing a swift decrease in size during subsequent convolutions, which may lead to the loss of critical information. The stride parameter defines the pixel advancement when the filter is applied to the image. In practical applications, a stride value of one or two is commonly utilized. The size of the output following convolution can be determined using the Eq. (15),

$$\frac{(W - F + 2P)}{S + 1} \tag{15}$$

In this context, W signifies the input size, F indicates the filter size, S represents the stride, and p denotes the amount of padding applied. When the input dimensions are specified as 5×5, with a filter size of 3×3, a stride of 2, and padding of 1, the calculated output dimensions will be 3×3. The Rectified Linear Unit (ReLU) is a commonly employed activation function used in conjunction with Convolutional Neural Networks (CNNs). The calculation is performed using the Eq. (16),

$$f(x) = max(0, x) \tag{16}$$

where a threshold of zero is used to exclude negative values. In addition to assisting with overfitting control, pooling layers decrease the input size, parameter count, and computational cost. Most pooling layers employ max pooling, which removes all except the highest-ranking elements from an input area.

Modeling of MR-CNN

One deep learning approach that can take use of MapReduce's parallelized computing is a Convolutional Neural Network. Upon the transmission of each training data instance through the network, CNNs iteratively update the network weights by computing their partial gradients. Sharing the data over several pieces allows for parallelization to take place during the training phase. It is therefore possible to train several CNNs simultaneously using each data block. After that, you may utilize the aggregated outcomes to change the weights for the following iteration. Hybrid MapReduce-CNN Structure is illustrated in Fig. 3.

Representation and CNN Architecture: Course and learner data are displayed as structured feature matrices, unlike image data. Each learner is represented as a row vector with behavioral features, such as rating history, time spent, and course completion rate. Each course includes topic embeddings, difficulty level, duration, and

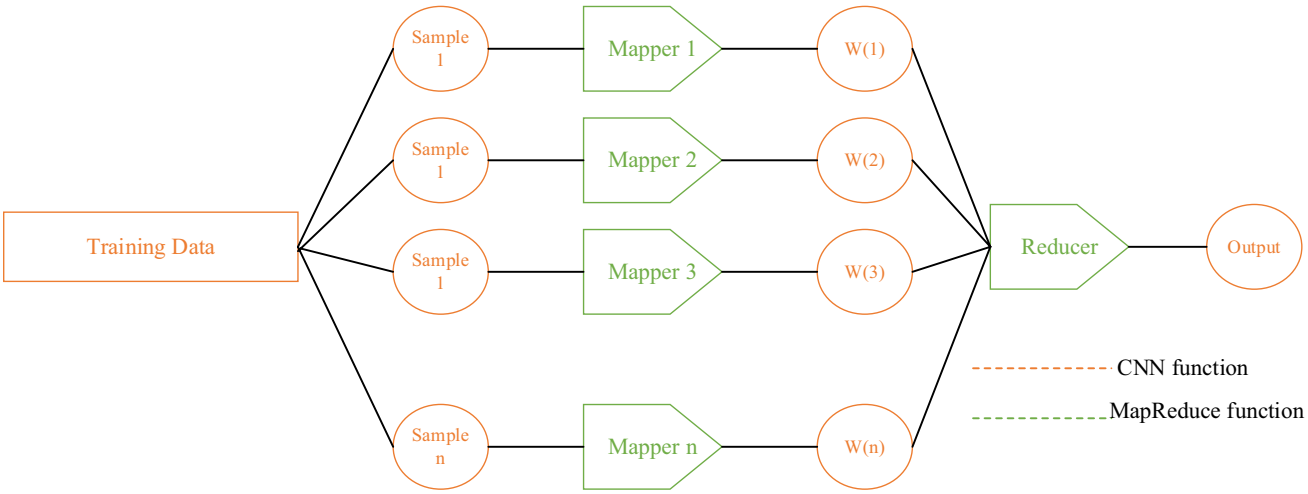


Fig. 3. Hybrid MapReduce-CNN structure.

instructor quality. These vectors create a 2-D feature grid suitable for convolution operations. The proposed CNN has four convolutional layers with 64, 128, 256, and 128 filters and a kernel size of 3 × 3. ReLU activation and 2 × 2 max-pooling follow each layer. A fully connected layer with 512 neurons combines the extracted features, with a dropout rate of 0.5 to avoid overfitting. The output layer uses a sigmoid function to calculate each course's relevance probability for a particular learner. This design helps the network capture both local and global relationships between learner preferences and course features.

```
The input to the mapper consists of a collection of tuples (x, y), where x represents a training sample and y denotes the corresponding ground truth label for that sample.
Weights is a list of randomly initialized weights for each hidden layer.
Function MAP() Read weights from HDFS;
Initialize CNN with random weights;
for i ← 0 to numLayers - 1 do
updatedWeights[i] ← updatedWeights[i] + Weights[i]
output: updatedWeights
```

Algorithm 1. Mapper function for MapReduce-Based CNN

The mappers receive an array of initialized weights [w1, w2,…, wn] along with the training data (x, y) as input parameters. The weights associated with hidden layer i are denoted as wi. This set comprises n weights, corresponding to the total number of hidden layers within the network. Tuples (x, y) containing training instances and their ground truth labels constitute the training data. Before training with input samples, each mapper sets the network's weights according to the provided specifications. Prior to being input into the reducer, the output is a collection of recently learned updated Weights [Δw1, Δw2,…, Δwn]. The mappers get a new set of weights after each iteration, and the network processes them until the maximum number of iterations is achieved.

The mappers' intermediate output is fed into the reducer, which then aggregates the weights generated by each mapper. Since mappers order weight positions in their outputs by hidden layer, the reducer may easily aggregate weights by index. In order to get an average of the weights, the aggregation first calculates the total weights and then divides it by the batch size of training examples. After that, the mapper receives the result and uses it to adjust the network's weights for the following cycle.

A driver is used by MapReduce jobs to schedule work. The driver breaks the program down into smaller jobs and generates an execution plan, often known as a directed acyclic graph (DAG). The driver interacts with Hadoop to configure jobs and find out which map and reduce classes are used. Paths to training data and output, as well as training parameters and network settings, are all part of the user-provided setups. The receptive field size, stride, number and type of layers, learning rate, and optimization method are all components of a network's design.

Purpose of MapReduce Integration: The MapReduce framework spreads CNN training across multiple nodes to handle large-scale e-learning data efficiently. Each mapper processes a separate part of the learner-course dataset, calculates partial gradient updates, and sends them to reducers, which gather and average these updates. This distributed learning method cuts down training time, balances the load, and allows for incremental retraining as new user interactions happen. Therefore, MapReduce improves the scalability and responsiveness of the MR-CNN recommendation model for real-time personalization.

The reducer receives input in the form of intermediate output generated by mappers, specifically the updated weights corresponding to each hidden layer (numLayers).
sumW eights is a list which stores the cumulative sum of weights
for each hidden layer
Function REDUCE()
sumWeights ← Initialize list of zeros;
for i ← 0 to numLayers-1 do
sumWeights[i] ← sumW eights[i] + W eights[i]
output: sumW eights

Algorithm 2. Reducer function for MapReduce-Based CNN

Mathematical formulation of evaluation metrics

Mathematical definitions for the key measures of quality are presented below in order to support an transparent analytical explanation for the quality of the model.

Precision@k computes the percentage of recommend courses that are relevant to the user and is computed using Eq. (17):

$$Precision@k = \left(\frac{1}{|U|}\right) \sum_{u \in U} \left[\frac{\left|R_u^k \cap G_u\right|}{k}\right] \tag{17}$$

where R_u^k stands for top-k recommendations for learner u, G_u is a set of relevant (ground-truth) courses, and |U| is the cardinality of learners.

Recall@k is the amount of successfully recommendered items that are pertinent which is shown in Eq. (18):

$$Recall@k = \left(\frac{1}{|U|}\right) \sum_{u \in U} \left[\frac{\left|R_u^k \cap G_u\right|}{|G_u|}\right] \tag{18}$$

Mean Average Precision (MAP@k) assesses ranking quality across students using Eq. (19):

$$MAP@k = \left(\frac{1}{|U|}\right) \sum_{u \in U} \left[\left(\frac{1}{|G_u|}\right) \sum_{i=1}^{k} P_{u(i)} \times rel_{u(i)}\right] \tag{19}$$

where $P_u(i)$ is the precision at cutoff i, and $rel_u(i)$ indicates whether the item at position i is relevant.

Normalized Discounted Cumulative Gain (NDCG@k) considers the salient items' ranking which is calculated using Eq. (20)

$$NDCG@k = \left(\frac{1}{|U|}\right) \sum_{u \in U} \left[\frac{DCG_u@k}{IDCG_u@k}\right] \tag{20}$$

where

$$DCG_u@k = \sum_{i=1}^{k} \frac{\left(2^{rel_{u(i)}} - 1\right)}{\log(i+1)} \tag{21}$$

and IDCGu@k is the ideal DCG obtained from perfectly ranked recommendations obtained from the Eq. (21).

Mean Absolute Error (MAE@k) evaluates the deviation between predicted and actual ratings using the Eq. (22):

$$MAE@k = \left(\frac{1}{|T|}\right) \sum_{(u,i) \in T_k} |\hat{r}_{ui} - r_{ui}| \tag{22}$$

and Root Mean Square Error(RMSE@k) can be calculated from the Eq. (23)

$$RMSE@k = \sqrt{\left[\left(\frac{1}{|T|}\right) \sum_{(u,i) \in T_k} (\hat{r}_{ui} - r_{ui})^2\right]} \tag{23}$$

where T_k is the top-k predicted rating set, and $\hat{r}_{ui}$ and r_{ui} are the predicted ratings and actual ratings, respectively. These mathematical expressions transparently and reproducably reveal the assessment of the performance. They

also allow comparison with the most up-to-date top-K recomendation models under standardized quantitative metrics.

Accuracy enhancement strategies

Accuracy of the presented recommendation model was enhanced by applying several integrated techniques. First, feature extension was applied by incorporating session-level learning behaviors and course content representations computed with TF–IDF and BERT modeling. These auxiliary features helped the system capture semantic structures and time-variant learning behaviors effectively. Second, regularization techniques such as dropout with a 0.5 rate and L2 weight decay with a $1 \times 10 - 5$ coefficient were applied to avoid overfitting and achieve enhanced generalization performance. Third, extensive hyperparameter tuning was performed with a grid search in learning rate, batch size, and filter settings to identify the best model parameters. Fourth, an ensemble fusion scheme was applied to combine the strengths of POA–Apriori and MR–CNN models by weighted averaging their outputs, leading to higher precision in recommendations. Fifth, time smoothing with a Kalman filter was applied to mitigate noisy engagement data and suppress fluctuations that can severely impact the learning process. Finally, methods of data augmentation were applied by synthesizing negative and mildly perturbed user–item relationships in order to enhance the robustness and flexibility of the model with unseen data.

Security and privacy considerations

Security To provide security and confidentiality for user information, the framework incorporates a series of protection measures at various stages of the data processing and recommendation flow. Transport Layer Security (TLS) is used to encrypt all data exchanges between server and client, and AES-256 is used to encrypt data that is kept in the system's data stores. Role-based access control is used to limit data access to approved individuals, and extensive audit logging is kept to log all access and edit activities. User information is also secured with anonymization and the stripping away of personally identifiable information (PII) prior to processing or analysis. Further privacy is enhanced during shared or federated learning activities by introducing differential privacy noise to shared updates of the model, lessening the potential for data reconstruction. The overall data handling process is GDPR-compliant and abides by institutional data ethics and consent procedures so that the system is secure, transparent, and in agreement with new privacy laws.

Reliability and fault tolerance

The reliability of the framework is maintained by a blend of architectural redundancy, model management, and monitoring techniques. Model version control and model checkpointing are enforced during training to maintain progress and provide recovery in the event of interruption. The deployment environment utilizes redundant servers and load-balanced inference endpoints to ensure continuous availability of service. Monitoring systems operate in real-time to capture latency, throughput, and accuracy drift, allowing anomaly detection prior to impacting the user experience. On detection of substantial drift in user behavior or data pattern, automatic retraining processes are initiated to reclaim model effectiveness. A backup collaborative-filtering recommender is also provisioned to act as a fall-back in the event of MR-CNN system unavailability, providing continuity in recommender delivery. These multifaceted measures improve the reliability, resilience, and fault tolerance of the overall recommender infrastructure.

Experimental analysis

Dataset detail

Udemy dataset

Information about courses offered by Udemy is included in the Udemy course dataset. Among the many features and factors included in the dataset are the following: cost, score, total number of users, number of reviews, course name, definition, teacher name, and over 130,000 courses. Many applications may make use of it, including data analysis, ML, RecSys, and natural language processing. Courses with the most subscribers, the greatest ratings, and the most popularity may be found with its help. For the purpose of developing strategies for the design and advertising of online courses, academics, investigators, and analysts can utilize this information to gain a better understanding of the Udemy platform and its users. There are 12,024 entries in the Udemy dataset. This dataset contains 3678 records from four different Udemy courses: web design, graphic design, business finance, and musical instruments. The massive open online course (MOOC) website Udemy offers both free and premium classes.

C. Coursera dataset

You can find details on all the courses offered by Coursera in their course dataset. There are almost 3900 courses in the collection, each with its own unique criteria such as length, language, rating, teacher, course name, description, and number of students enrolled. You may use the dataset for data analysis, ML, RecSys, and natural language processing; it is freely downloadable (3). It can provide useful data on the most popular, ranked, and enrolled courses. Students, teachers, and researchers may all utilize this information to learn more about Coursera and its users, as well as to plan for the future of online education. Additions of new courses and updates to the platform cause regular updates to the Coursera course dataset. The dataset in question was culled from publicly accessible Coursera data in September 2021 and, in the event that any data extraction errors occurred, was subsequently filled up by hand. Depending on the level of difficulty and the skills required, it may be utilized in RecSys to promote Coursera courses. Dataset of courses extracted from the Coursera platform. The dataset contains 3522 course data points organized into six columns. Table 2 displays the experimental setup and parameter values used in the model validation experiment.

For both Udemy and Coursera datasets, model training features included course ratings, enrollment numbers, duration, topic vectors, learner demographics, and engagement statistics. Each dataset was split into training (80%) and testing (20%) sets while ensuring user independence between splits to avoid information leakage. All numerical attributes were standardized to have a zero mean and unit variance.

Validation protocol

To guarantee the reliability and validity of the introduced model, a stringent validation framework was adopted. The data were split into an 80/20 train–test split, with user independence maintained to avoid data leakage during training and test phases. Furthermore, five-fold cross-validation was also used on training data to optimize hyperparameters and assess the consistency of the model in multiple splits. In statistical validation, paired t-tests with a significance level $\alpha = 0.05$ were also performed to compare the introduced MR–CNN scheme with baseline schemes, with a guarantee that improvement was statistically significant. Also, an ablation study was performed to quantify the contribution of each component like the Kalman filter and POA–Apriori optimisation and show their effect on overall performance. Moreover, bootstrap analysis with 1000 resamples was also computed to estimate 95% confidence intervals for crucial evaluation metrics with robustness in estimating stability in the model. In entirety, these procedures confirm the statistical soundness, generality, and reliability of the introduced system's findings.

Performance analysis

A course was deemed relevant if a learner enrolled in it or rated it 4 out of 5 or higher. Performance was assessed for top-K recommendations (K = 5, 10, 15, 20). Each experiment was run five times with different random splits, and results were reported as mean ± standard deviation. Statistical significance between the proposed model and baselines was confirmed using a paired t-test. The strong performance of the MR-CNN model comes from its ability to learn in a distributed manner and extract features in a hierarchical way. The MapReduce layer speeds up training by using data partitions, which helps reduce overfitting on large datasets. At the same time, the CNN captures both the overall and specific relationships between users and courses. The POA-Apriori module improves the accuracy of rules by adjusting confidence thresholds, resulting in better associations between learners and courses. These features explain the improvements seen in precision, recall, and NDCG metrics.

Precision@k is a metric that evaluates the alignment of the top k recommended courses with the interests and requirements of the student; it serves as an assessment statistic. The objective is to assess the accuracy of a recommendation system (RecSys). When a student perceives value, enjoyment, or engagement in a course, it is classified as relevant. The system is assessed at k = 5, where precision@k is calculated as 0.4. In this scenario, RecSys recommends 10 courses to a student, of which 6 are pertinent to the student's requirements. The rationale is that, within the top five suggestions, there were six pertinent courses identified. Considering the relevance of the courses and the order of recommendations, Precision@k serves as a valuable metric for evaluating the leading suggestions generated by the proposed PCR system.

Recall@k measures the proportion of the top k courses that have been assessed as relevant. When a student engages with or finds a proposed course fascinating, it is classified as relevant. In a scenario where a student has participated in 10 courses and the recommendation system proposes 5 courses, with 3 of those being pertinent to the student, the Recall@5 metric would be calculated as 0.3.

MAP@k determines the suitable course proportion for each student through an analysis of the top k recommended courses, subsequently calculating the average of these proportions. For the purpose of illustration, consider that 80% of the pertinent courses for each student are included within the top 5 recommended courses when the recommendation system is evaluated at k = 5, resulting in a MAP@k value of 0.8. In a situation where students lack knowledge of the total number of pertinent courses, MAP@k assesses the proposed PCR system by analyzing the recall and precision metrics of the recommendation system. The issue of distinguishing between relevant courses that are ranked higher or lower in the recommendation list is addressed through metrics such as NDCG.

NDCG@k is employed to evaluate the effectiveness of recommendation systems by measuring the quality of course recommendation rankings. A normalized variant of.

Discounted Cumulative Gain (DCG@k), NDCG@k employs a logarithmic discount applied to the placements of the courses to calculate the aggregate relevance ratings for the top k suggested courses. Courses with superior rankings are prioritized through the application of a logarithmic discount. The calculation of NDCG@k involves dividing DCG@k by the ideal DCG@k. The ideal DCG@k is determined by prioritizing all relevant courses. An NDCG@k value of 0.9 signifies that, on average, the ranked suggested courses achieve 90% of the quality of the ideal ranking, specifically for the RecSys evaluation at k = 5.

RMSE@k calculates the mean value for all courses in the test set, assessing the deviation of actual ratings from the expected ratings. The metric is calculated by computing the square root of the mean of the squared differences between the actual ratings and the expected ratings. To calculate RMSE@k for recommendations, only the top k courses are considered. The RMSE is computed solely based on the actual and predicted ratings of the top k courses. When the RecSys is evaluated at k = 10 and the RMSE@10 is recorded as 1.5, this indicates that, on average, there exists a 1.5 unit difference between the predicted ratings and the actual ratings for the top ten recommended courses.

MAE@k calculates the mean absolute deviation between the expected ratings and the actual ratings of all courses in the test set. The study identifies the top k courses within the recommended catalog and utilizes them to calculate MAE@k. The MAE is computed by utilizing only the anticipated and actual ratings of the top-k courses. The average discrepancy between the projected ratings of the top ten suggested courses and the actual ratings is 1.2 units, indicated by an MAE@10 of 1.2 during the evaluation of a RecSys at k = 10.

The Mean Squared Error at k (MSE@k) must be computed for the leading k suggestions. MSE prioritizes the correctness of the most significant suggestions over the consideration of all proposals. MSE@k serves as an effective metric for assessing the performance of recommendation systems, particularly when the objective is to deliver a concise collection of high-quality recommendations to the student. The system demonstrates proficiency in predicting the student's preferences for the top 'k' suggestions, as indicated by a low MSE@k. This capability contributes to enhanced user experience and satisfaction.

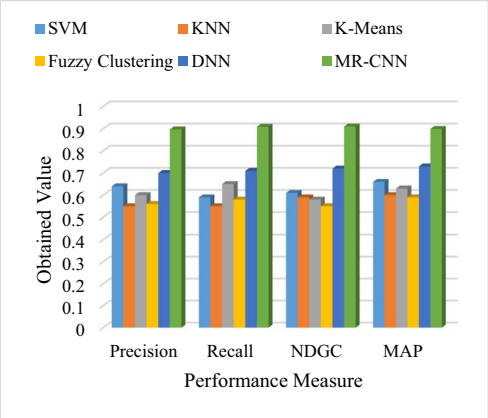
The baseline models—SVM, KNN, K-Means, Fuzzy Clustering, and DNN—were chosen to represent different types of recommendation approaches: traditional classification (SVM, KNN), unsupervised grouping (K-Means, Fuzzy C-Means), and deep learning (DNN). These models offer a fair comparison across linear, similarity-based, and neural methods. All baseline parameters were adjusted using grid search, such as setting the SVM kernel to RBF, KNN neighbors to 10, and the learning rate to 0.001 for DNN.

As shown in Table 3, the MR-CNN model consistently achieves the highest precision, recall, and NDCG scores at all K levels. This shows its effectiveness in capturing learner preferences. The decrease in MAE and RMSE indicates that the model can predict user ratings more accurately. Traditional methods like SVM and KNN fall short because of their linear assumptions. In contrast, deep models such as DNN lack the efficiency of distributed learning. The hybrid architecture balances scalability and depth, leading to ongoing improvements across evaluation metrics.

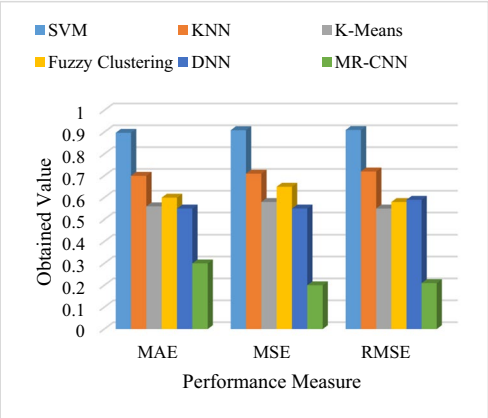
Figure 4 shows the Comparative analysis for different measures. What follows is a discussion of the experimental outcomes. Figure 4 provides a comparison of all models. The MR-CNN framework shows better accuracy and ranking quality on all evaluation measures. Unlike traditional methods like MF and NCF, which have trouble with sparse learner-course matrices, MR-CNN successfully combines distributed feature extraction with optimized association rules. The findings reveal that while increasing the number of predictive factors initially boosts performance, overfitting later limits these gains. Therefore, careful adjustment of the convolutional depth and Apriori thresholds is crucial for achieving the best results. In addition, we discovered that the two measurements show the same trend of change. Outperforming the MF and NCF methods, the DÉCOR model showed the best performance when there were 32 predictive elements. Also, after NCF and ItemKNN came MF, and MF did somewhat better than NCF. Even though MF's performance dropped when the number of predictive factors was large due to overfitting, when the number of factors was smaller, MF outperformed NCF.

Metrics	Number of top recommendations *k*	Models					
		SVM	KNN	K-means	Fuzzy clustering	DNN	Proposed MR-CNN
Precision@k (higher is better)	5	0.6123	0.5479	0.7868	0.5678	0.6890	0.8908
	10	0.5876	0.5690	0.7678	0.5908	0.7890	0.8769
	15	0.5908	0.5368	0.7679	0.5678	0.7654	0.8689
	20	0.6576	0.6269	0.7845	0.6890	0.7678	0.8867
Recall@k (higher is better)	5	0.5647	0.5257	0.7856	0.6456	0.7450	0.8974
	10	0.6578	0.6480	0.7045	0.6689	0.7578	0.8896
	15	0.6960	0.6357	0.7890	0.6969	0.7698	0.8690
	20	0.6478	0.7289	0.7845	0.6468	0.6799	0.8907
NDCG@k (higher is better)	5	0.6790	0.6148	0.7245	0.6789	0.7458	0.8967
	10	0.6468	0.7649	0.7568	0.7500	0.7680	0.8765
	15	0.7170	0.7365	0.7357	0.7679	0.7753	0.86967
	20	0.7148	0.7258	0.7956	0.7580	0.7636	0.8567
MAP@k (higher is better)	5	0.6538	0.6257	0.7904	0.6570	0.7459	0.8648
	10	0.6940	0.7257	0.7345	0.6860	0.7869	0.8678
	15	0.7529	0.7479	0.7567	0.6868	0.7867	0.8759
	20	0.7435	0.7479	0.7456	0.7567	0.8907	0.8648
MAE@k (lower is better)	5	0.4379	0.4579	0.3567	0.4876	0.4567	0.3468
	10	0.4863	0.4689	0.3678	0.4680	0.4101	0.2550
	15	0.4648	0.4379	0.3908	0.4789	0.4789	0.2590
	20	0.4864	0.4763	0.3876	0.4500	0.4567	0.2700
MSE@k (lower is better)	5	0.4742	0.4268	0.3765	0.4670	0.4967	0.2456
	10	0.4905	0.4623	0.3876	0.4680	0.4789	0.2567
	15	0.4742	0.4578	0.3978	0.4789	0.4579	0.2456
	20	0.4260	0.4246	0.2459	0.4509	0.4890	0.6789
RMSE@k (lower is better)	5	0.5479	0.4846	0.39788	0.4780	0.3678	0.3578
	10	0.5680	0.4579	0.3567	0.4540	0.3879	0.3568
	15	0.5146	0.4856	0.3356	0.4680	0.3456	0.3689
	20	0.5279	0.4576	0.3856	0.4679	0.3879	0.3456

Table 3. Performance of all the recommendation models for comparison on the Udemy dataset: bold indicates the best performance value in the metric in a comparison among proposed models.



a) Precision, Recall, NDGC and MAP (Proposed MR-CNN Shows improved results)



b) MAE, MSE & RMSE (Proposed MR-CNN Shows lower error rates)

Fig. 4. Comparative performance of proposed MR–CNN and baseline models across Precision@k, Recall@k, NDCG@k, MAE@k, and RMSE@k metrics.

The outcomes demonstrate that the DÉCOR model demonstrated strong performance when combining the UBE layer, which records user behavior attributes, with the CAE layer, which records course attribute attributes. Secondly, we discovered that the performance was enhanced by adding more predictors. A total of 32 predicting criteria yielded the optimal DÉCOR performance. The latent factor-based model's capture power also grew as the predictive factor did. At the same time, plenty of noise was present, the model experienced issues with overfitting, and performance began to decline. In order to get the most out of the model, we must determine how many predictive factors to use[38].

Conclusion

The implementation of advanced methodologies and state-of-the-art technology facilitates the creation of a smart learning environment that enhances the growth of personalized and adaptable education. In line with this trend towards integration, there has been a proliferation of IoT-enabled smart education systems. These systems have the potential to provide students with Personalized Course Recommendations (PCR) by utilizing Machine Learning (ML) approaches. Present recommendation algorithms often fail to take advantage of chances to capture learners' changing preferences due to their reliance on either explicit or implicit feedback, even when they integrate both types of information. Using an intelligent E-Learning (EL) platform, we present a new method for PCR and personalized learning in this research. The model will take into account each student's interests, learning style, and academic performance to figure out which courses would be the best fit. Our method suggests courses based on the learner's interactions with the system and the cosine similarity of related content by merging explicit (user ratings) and implicit (views and behavior) methods. The initial step involves preprocessing the data to enhance accuracy and reduce noise levels. The subsequent phase involves utilizing a hybrid approach that integrates the Pufferfish Optimization Algorithm with the Apriori method (POA-Apriori) for the purpose of mining user interests. In conclusion, a hybrid Map-Reduce based Convolutional Neural Network (MR-CNN) was proposed to enhance the efficiency of recommendation generation. Two benchmark datasets are utilized for the experiments: one sourced from Coursera and the other from Udemy. The proposed model demonstrates superior performance compared to existing top-K recommendation methods when evaluated using accuracy measures including PCR's Mean Absolute Error (MAE)@k, Root Mean Squared Error (RMSE)@k, Normalized

Discounted Cumulative Gain (NDCG)@k, and precision@k. Superior recall, MAP, and NDCG values, as well as reduced MAE, RMSE, and MSE values, distinguish MR-CNN from the other proposed methods.

The experimental findings show that the hybrid POA, Apriori, and MR-CNN model improves personalization and scalability in distance education recommendation systems. These results highlight the model's ability to combine optimization, association mining, and distributed deep learning to understand complex learner-course relationships. However, like many AI-driven frameworks, this study has some limitations. The datasets used, Udemy and Coursera, are limited to specific areas and may not capture the full variety of online learning environments. Also, the model's computational cost increases with data size because of multi-layer convolution operations, which may hinder real-time deployment on low-resource systems. Future work will aim to tackle these limitations by looking at lightweight or transformer-based architectures for quicker and more efficient personalization. Adding explainable AI components will improve transparency and clarity in course recommendations. Further studies will also focus on incorporating multimodal learner data, including behavioral logs, eye-tracking, and voice interactions, to improve adaptive learning pathways. Expanding the system to other educational fields and implementing real-time feedback loops could enhance the overall flexibility and effectiveness of the proposed model.

Author contributions

Xia Yang—Writing—original draft, Writing—review & editing, Methodology, Project administration.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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References

1. Chi, C. et al. An intelligent recommendation system for personalised parametric garment patterns by integrating designer's knowledge and 3D body measurements. *Ergonomics* **68**, 1–21 (2024).
2. Zhou, R. et al. Research on grid data analysis and intelligent recommendation system by introducing neural tensor network model. *Scalable Comput. Pract. Exp.* **25**, 3996–4004 (2024).
3. Xin, X et al. Knowledge-based intelligent education recommendation system with IoT networks. *Secur. Commun. Netw.* (2022).
4. Gajbhiye, S. et al. Higher education recommendation system. *Int. Res. J. Modernization Eng. Technol. Sci.* (2023).
5. Student, Janagani Nithin Reddy et al. *Higher Education Recommendation System Using XG-Boost Algorithm*. (2023).
6. Falola, P. B. et al. Development of an android-based reminder system for social services. *Proc. Comput. Sci.* **258**, 487–496. https://doi.org/10.1016/j.procs.2025.04.284 (2025).
7. Wang, C., Akinlade, O. & Ajagbe, S. A. Dynamic resilience assessment of urban traffic systems based on integrated deep learning. In *Advances in Transdisciplinary Engineering*, 70: Intelligent Transportation and Smart Cities. **70**, pp. 33–42. (2025). https://doi.org/10.3233/ATDE250238
8. Communication Networks, Security and. Retracted: Knowledge-based intelligent education recommendation system with IoT networks. *Secur. Commun. Networks.* (2022).
9. G M, Dhananjaya et al. "A digital recommendation system for personalized learning to enhance online education: A review. In *IEEE Access.* **12** (2024), pp. 34019–34041.
10. Fedushko, S. S. et al. Intelligent academic specialties selection in higher education for Ukrainian entrants: A recommendation system. *J. Intell.* **10**, 32 (2022).
11. Xu, S. & Yin, X. Recommendation system for privacy-preserving education technologies. *Comput. Intell. Neurosci.* **202**, 3502992 (2022).
12. Yang, T. & Zhiqun, D. Design of personalized ideological and political education resource recommendation system in the online education platform in the network environment. *J. Environ. Public Health.* **2022** (2022).
13. Vo, N. N. Y. et al. CRS—A hybrid course recommendation system for software engineering education. In *2022 IEEE/ACM 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)*. pp. 62–68 (2022).
14. Fu, R. et al. The design of personalized education resource recommendation system under big data. *Comput. Intell. Neurosci.* **2022**, 1359730 (2022).
15. Baidada, M. et al. Hybrid filtering recommendation system in an educational context: Experiment in higher education in Morocco. *Int. J. Web Based Learn. Teach. Technol.* **17**, 1–17 (2022).
16. Saito, T. & Watanobe, Y. Learning path recommendation system for programming education based on neural networks. *Int. J. Distance Educ. Technol.* **18**, 36–64 (2020).
17. Zhang, R. A personalized course resource recommendation method based on deep learning in an online multi-modal multimedia education cloud platform. *Int. J. Inf. Technol. Syst. Approach* **16**, 1–14 (2023).
18. Li, S. & Yang, Bo. Personalized education resource recommendation method based on deep learning in intelligent educational robot environments. *Int. J. Inf. Technol. Syst. Approach* **16**, 1–15 (2023).
19. Karabila, I. et al. Recommendation system using deep learning-based sentiment analysis. In *2023 Sixth International Conference on Vocational Education and Electrical Engineering (ICVEE)*. pp. 41–47 (2023).
20. Lin, T. et al. Deep learning-based course recommendations using sentence embeddings and user information for learning platforms. In *2024 28th International Computer Science and Engineering Conference (ICSEC)*. pp. 1–6 (2024).
21. Chen, Y. et al. An empirical systematic study of deep learning-based software defect prediction models in complex systems. In *2024 International Conference on Computers, Information Processing and Advanced Education (CIPAE). pp.* 43–48 (2024).
22. Lei, L. Research on personalized education recommendation algorithm based on artificial intelligence. In *2023 International Conference on Applied Physics and Computing (ICAPC)*. pp. 531–535 (2023).
23. Yang1, M. Design of personalized recommendation system for college education based on multivariate hybrid criteria fuzzy algorithm. *J. Electr. Syst.* (2024).
24. Chen, C. & Huang, J. Personalized learning scheme of tourism management professional curriculum system in colleges and universities under intelligent recommendation system based on OBE education model. *Appl. Math. Nonlinear Sci.* **9**, 20242243 (2024).
25. Wen, L. & Geng. J. Research on personalized early childhood education recommendation model based on improved DeepFM. In *Proc. of the 2023 4th International Conference on Computer Science and Management Technology.* (2023).

26. Lee, Y et al. "Personalized Recommendation System based on Recommendation Indicator in Education Platform. In *Proc. of the 2023 International Conference on Research in Adaptive and Convergent Systems*. (2023).
27. Qu, Z. Personalized recommendation system for english teaching resources in colleges and universities based on collaborative recommendation. *Appl. Math. Nonlinear Sci.* **9**, amns-2024 (2024).
28. Moon, K. Z. et al. BERT-based personalized course recommendation system from online learning platform. In *2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*. pp. 980–985 (2024).
29. Xie, F. Intelligent personalized recommendation method based on optimized collaborative filtering algorithm in primary and secondary education resource system. *IEEE Access* **12**, 28860–28872 (2024).
30. Duong Tranh Tran, P. et al. Personalized learning paths recommendation system with collaborative filtering and content-based approaches. *Sci. Technol. Dev. J. Econ. Law Manag.* **8**, 5243–5253 (2024).
31. Yanyan, Z. Artificial intelligent recommendation system for ideological and political education content in photoshop course. In *2024 IEEE 4th International Conference on Electronic Communications, Internet of Things and Big Data (ICEIB)*. pp. 286–290. (2024).
32. Surendran. R. et al. Friendly course equalizing cloud service using education cloud. In *2021 International Conference on Computing, Electronics & Communications Engineering (iCCECE)*, Southend, United Kingdom. pp. 7–12. (2021). https://doi.org/10.1109/iCCECE52344.2021.9534845.
33. Krishnan, S., Rajendran, S. & Zakariah, M. Secured accreditation and equivalency certification using Merkle mountain range and transformer based deep learning model for the education ecosystem. *Sci. Rep.* **15**, 22511. https://doi.org/10.1038/s41598-025-06789-x (2025).
34. Joypriyanka, M., Surendran. R., Sathish. J. & Sivasangari, A. Game therapy for specially-abled individuals with PPO reinforcement learning in VR-based educational games. In *2024 5th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, Tirunelveli, India. pp. 1126–1133. (2024) https://doi.org/10.1109/ICDICI62993.2024.10810847.
35. Krishnan, S. & Surendran, R. Impact of equivalence assessment in the education sector using the XAI model of blockchain with ECTS. In *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, Salem, India, pp. 13–20. (2024) https://doi.org/10.1109/ICAAIC60222.2024.10575529.
36. De Santo, A. Semi-supervised and deep learning approaches to social network community analysis. *Knowl. Based Syst.* **216**, 107345 (2021).
37. Daneshfar, F., Dolati, M. & Sulaimany, S. Graph Clustering Techniques for Community Detection in Social Networks. In *Community Structure Analysis from Social Networks*, 1st ed., Chapman and Hall/CRC. pp. 1–20. (2025).
38. Wang, N. Ideological and political education recommendation system based on AHP and improved collaborative filtering algorithm. *Sci. Progr.* **2021** (2021), 2648352:1–2648352:9. (2021).

Author contributions

Xia Yang - Writing- original draft, Writing- review & editing, Methodology, Project administration

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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